COMP306 – Database Systems

The Starlight Carnival Management System

Business Scenario

A discussion into the managerial workings of an Amusement Park, defining the entities and relationships as well as their responsibilities for daily Operations.

by Group G

University of KwaZulu-Natal



Project Formalities

Amusement Park Management System

The Starlight Carnival

Group G

Group Members

Member Name	Student ID
Jawad Kachbal	223037045
Joshua Blom	224039813
Mahomed Valli Mahomed	224005878
Abiel Govender	224058851
Andiswa Hlengwa	221002430

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Business Scenario: The Starlight Carnival

The Starlight carnival is a wonderous travelling amusement park that travels the lands, bringing entertainment, thrills and family fun to communities across the continent. The park's operations are managed from a centralized mobile headquarters, ensuring that all daily activities are recorded and tracked through a comprehensive database management system. The system is designed to handle customer admissions, employee roles, attraction operations, maintenance scheduling, inventory management and financial transactions during their temporary stay at various **Location**s. Each location is characterized by its city, venue name, physical address, arrival and departure dates as well as the rental cost for the duration of the stay. **A single location hosts many admissions, many transactions, many employees, and many attractions.** These location records allow management to keep track of performance metrics during the carnival's annual tour, allowing them to plan future stops based on profitability and logistical feasibility.

The Starlight Carnival offers its visiting customers the opportunity to sign up for a Starlight Smart Card, serving as a loyalty membership card and a rechargeable payment method that can be used across the Carnival. To register for a smart card, customers need to provide their name, surname, email address, phone number and date of birth. The system will automatically calculate the age group classification of Child, Adult or Senior based on the DOB provided. Upon successful registration, a Customer record is created containing the customers personal details, signup date, current card balance, accumulated loyalty points and card status. The smart card balance can be topped up at any ticket booth or kiosk within the park, with funds being able to be used to pay for entry fees, use at stalls and premium access passes for rides and shows. **A single customer may have many admissions over multiple visits, may conduct many transactions, and may purchase many access passes.** Customers who choose not to register for a smart card may still enter the park by paying the standard entry fee. In this case, an **Admission** record is created without an associated customer identifier and have the option to use standard payment methods (Cash or Card) within the carnival.

Customer Operations

Upon arrival at the park entrance, guests are asked if they possess a Starlight Smart Card. If they do, the card is scanned to verify active status and sufficient balance. An **Admission** record is created linking the entry to their customer profile and current location. If they do not have a card, they have the option to register for one or pay the entry fee as a guest. The admission record captures the entry date and time, the admission type based on the guest's age group, the entry fee paid and later records the exit time when the guest leaves the park.

All monetary exchanges such as entry fees, smart card top-ups and stall purchases are recorded as **Transaction** records. Each transaction is linked to the customer if the a smart card was used, the employee who processed the transaction and the location where the transition occurred. **A single employee processes many transactions during a shift.** Transactions are timestamped and categorised by type, with each one being settled through one or more **Payment** records, specifying the tender type (Cash, Card, Smart Card Balance). **A single transaction may be settled through one or more payments.**

To enhance the guest experience, The Starlight Carnival offers premium **AccessPass** options that allow guests to bypass standard queues or secure reserved seating. The **AccessPass** supertype captures the transaction through which the pass was purchased, the customer who purchased it, the price paid, time of issue and whether the pass has been redeemed. Access passes are divided into two subtypes. **Each access pass must be exactly one subtype and cannot belong to both simultaneously.** A **RidePass** is associated with a specific ride and grants the holder a designated time slot during which they may board the ride with minimal waiting. A **ShowReservation** is linked to a particular show and guarantees the holder specific seat number for a scheduled performance time. **A single transaction may include multiple access passes.** These passes are scanned at the attraction entrance by employees to validate the redemption and prevent fraudulent reuse.

Internal Operations

The Starlight Carnival employs a diverse workforce whose members are all recorded in the **Employee** Entity, serving as the supertype which contains shared attributes such as, name, surname, hire date, hourly wage and current assigned location. Every employee is further classified into one of four specialised subtypes. **Every employee must be classified into exactly one subtype, and no employee may belong to more than one subtype. Operators** are responsible for running the mechanical rides and are assigned a ride responsibility category which indicates the type of ride they operate based on the intensity of the ride: Thrills for high intensity, Kiddies for Low intensity and Motion for Moderate intensity rides. **Technicians** are skilled maintenance workers who hold specialisations in areas such as hydraulics, electrical systems and General mechanics. Each technician is assigned a unique toolbox identifier and an emergency contact number for afterhours incidents. **A single technician performs many maintenance jobs over time. Vendors** operate the various stalls throughout the park and are assigned to a specific stall location for their shift. **Each stall is assigned exactly one vendor at any given time. Performers** are the entertainers who bring the carnival's shows to life with each performer being assigned to a specific show in which they appear. **Each performer is assigned to exactly one show, though a single show may feature many performers.** The final subtype is **GeneralStaff**. This encompasses all other employees such as those in sales, managerial positions, cleaning and sanitation as well as security personnel. This categorization allows The Starlight Carnival to keep track of its staff activities accordingly.

The attractions at The Starlight Carnival form the core of its experience and are represented by the **Attraction** supertype. Every attraction has a name, an operational status that indicates whether it is currently operational, closed temporarily or undergoing maintenance, and a location where the attraction has been setup. Attractions are further divided into three subtypes. **Every attraction must be categorized into exactly one of the three subtypes. Rides** are mechanical amusements categorised by type (Thrills, Kiddies, Motion), with each ride enforcing an age requirement specifying whether it is suitable for Adults only, Kids Only or all Ages. **A single ride accumulates many maintenance records throughout its operational lifetime. Shows** are live performances characterised by their genre, duration in minutes, indoor or outdoor and their seating capacity. **Stalls** are booths identified by a lot number that indicates their physical position on the carnival grounds. Each stall has is assigned a type (Food, Merchandise and Games). Vendors are assigned to specific stalls, with each stall maintaining an inventory of Products that are sold to guests. **A single stall sells many products.**

Maintaining the safety and reliability of mechanical rides is a critical operational function. All maintenance activities are documented in the **MaintenanceLog** entity. Each log entry records which ride was serviced, which technician performed the work, the date and time of the maintenance, a detailed description of the issue found, action taken and the cost of any replacement parts used. This allows management to keep track of upkeep and assess technician performance.

The Starlight Carnival deploys a comprehensive management system. Management can query the database to compare revenue generated across different locations, analyze which attractions are most popular among different age groups, identify top performing vendors and products, track maintenance costs per a ride and monitor customer engagement. By integrating customer admissions, employee management, attraction operations, inventory control and financial transactions into a single unified database, The Starlight Carnival ensures smooth daily operations whether the park is stationed in a bustling metropolitan center, or a quiet regional town.

AI Usage

- Plausibility checking
- Asstance with polishing Entity details
- Grammar Checking